

iLumos Claim Chart Refinement MVP

Product requirements document · Assessment prototype

MATTER US123456 vs. Acme Corp Thermostat · STATUS Working draft

Problem statement

Patent analysts already create claim charts, but strengthening the reasoning and evidence behind each claim element is slow and difficult to audit. Generic AI chat can also overstate what a source proves. iLumos makes refinement faster while keeping the analyst responsible for the legal conclusion: every suggestion is source-grounded, visibly provisional, and applied only after human review.

User stories

- As a patent analyst, I want to upload a claim chart and product documents so the AI works from the same evidence I am reviewing.
- As a patent analyst, I want to request a refinement against a specific claim element so the response is precise rather than generic.
- As a patent analyst, I want to accept, reject, edit, or undo a suggestion so the chart never changes without my control.
- As a patent analyst, I want the AI to identify evidence gaps instead of inventing support so I know what to research next.

Core Features

IN SCOPE	OUT OF SCOPE
Document upload and source list; matter-level instructions; three-column chart; element-focused chat; source-grounded suggestions; evidence status; accept, reject, edit, undo; missing-evidence request; Word-compatible export.	Authentication and permissions; production-grade parsing; live web scraping; automated infringement conclusions; collaboration; full version management; legal advice.

Key decisions

- Keep the chart and chat side by side: the analyst can see exactly which row is in context and verify the result without losing the source mapping.
- Treat AI output as a proposal: suggestions are never written automatically; accept, reject, edit, and undo make the human-in-the-loop boundary explicit.
- Separate disclosed fact from inference: citations are required, invented evidence is prohibited, and qualified language is used when a source suggests rather than proves an implementation.

Acceptance criteria

- Analysts can add a claim chart and product documents, then save matter-level instructions.
- Interface displays the three chart columns and lets the analyst focus a specific element.
- A refinement request returns a specific, source-labelled suggestion for the focused element.
- Analysts can accept, reject, or edit a suggestion; accepted changes are visible in the chart.
- Analysts can undo the last applied refinement and restore the prior chart state.
- Insufficient evidence produces a request for a document or URL and does not strengthen the conclusion.
- Refined charts can be exported for continued work in Word.

Success metrics

Median time to move an element from needs review to an evidence-backed draft; suggestion decision rate; percentage of refined rows with an identifiable source; unsupported-claim rate found in analyst review; export completion rate without unresolved required fields.

Assumptions

The initial experience is for one analyst in one matter. The assessment prototype uses the supplied Acme thermostat example and deterministic demo responses; production would connect the same interaction model to retrieval, document parsing, source citations, and audit-grade versioning.

~ PRD written by Vivek for Lumenci.